The empirical recurrence for OEIS A275092, recovered from its tail

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8 September 2026

Abstract

OEIS A275092 records "Empirical recurrence of order 84" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1223$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{84} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A275092 is "Number of 4Xn 0..2 arrays with no element equal to any value at offset (0,-2) (-1,-2) or (-2,-1) and new values introduced in order 0..2.". It has offset 1 and begins

$$14, 486, 1024, 3088, 12816, 53132, 232156, 984288, \ \ldots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Jul 16 2016 09:25:01, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1223$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1223$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2446$ exact terms from $n = 5$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{84} c_i \, a(n-i),$$

c_1	2	c_{22}	48044376	c_{43}	−1822640662	c_{64}	−11511548
c_2	25	c_{23}	−67448470	c_{44}	−1228989816	c_{65}	−4435551
c_3	−22	c_{24}	−93545358	c_{45}	1553285819	c_{66}	2974586
c_4	−272	c_{25}	123793227	c_{46}	1260280890	c_{67}	1925242
c_5	−59	c_{26}	155211087	c_{47}	−1127709676	c_{68}	−534736
c_6	2082	c_{27}	−175506442	c_{48}	−1144796149	c_{69}	−592522
c_7	3041	c_{28}	−189574049	c_{49}	647589221	c_{70}	36787
c_8	−14800	c_{29}	217383722	c_{50}	871689163	c_{71}	150789
c_9	−26795	c_{30}	135957251	c_{51}	−292065021	c_{72}	7803
c_{10}	82545	c_{31}	−316394908	c_{52}	−535287307	c_{73}	−26862
c_{11}	121306	c_{32}	−38288279	c_{53}	129722585	c_{74}	−1641
c_{12}	−323928	c_{33}	591349131	c_{54}	294196151	c_{75}	4325
c_{13}	−321979	c_{34}	38792280	c_{55}	−87826852	c_{76}	−164
c_{14}	1033800	c_{35}	−1093843124	c_{56}	−174314857	c_{77}	−1191
c_{15}	288527	c_{36}	−184920274	c_{57}	57719878	c_{78}	−25
c_{16}	−3080971	c_{37}	1601818122	c_{58}	114252192	c_{79}	271
c_{17}	1453191	c_{38}	482075985	c_{59}	−21571726	c_{80}	37
c_{18}	8789241	c_{39}	−1874342866	c_{60}	−69311631	c_{81}	−34
c_{19}	−8649909	c_{40}	−832248276	c_{61}	382700	c_{82}	−10
c_{20}	−22053913	c_{41}	1936258325	c_{62}	32896224	c_{83}	2
c_{21}	28155966	c_{42}	1078751499	c_{63}	5723279	c_{84}	1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A275092 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \cdots - c_{84}$. Its constant term is $-c_{84} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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